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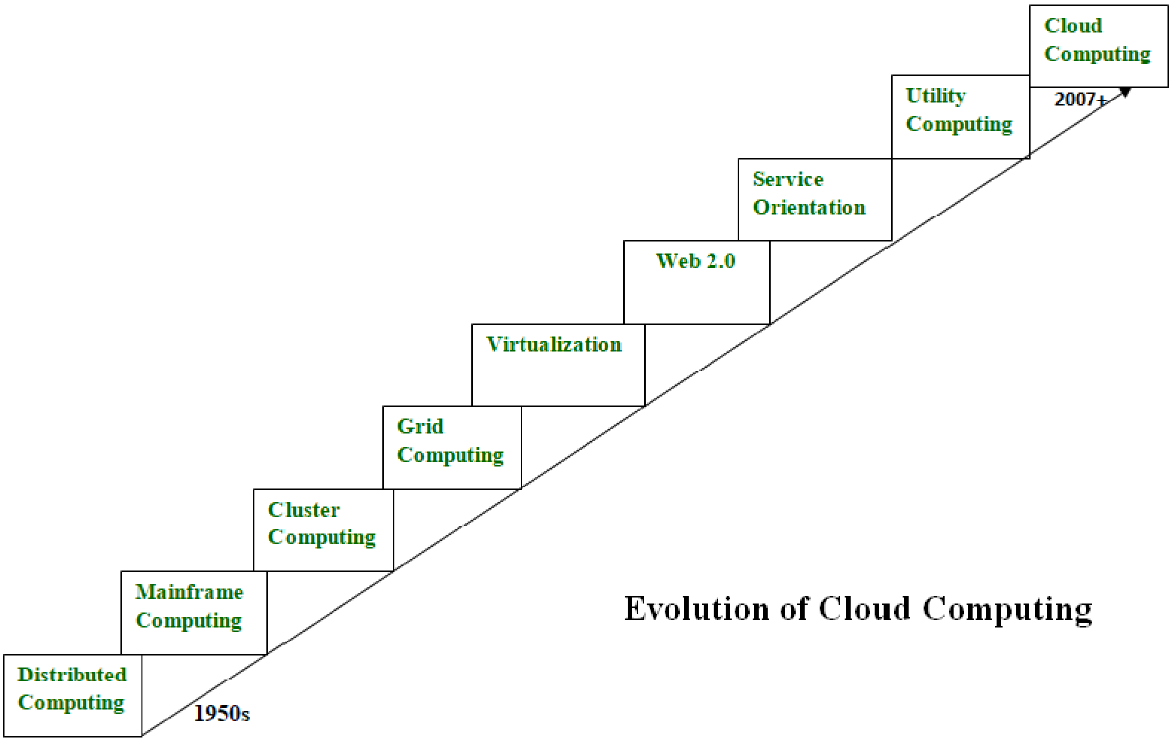
Evolution of Cloud Computing

Cloud computing allows users to access a wide range of services stored in the cloud or on the Internet. **Cloud computing** services include computer resources, data storage, apps, servers, development tools, and **networking protocols**. It is most commonly used by IT companies and for business purposes.

Evolution of Cloud Computing

The phrase "Cloud Computing" was first introduced in the 1950s to describe internet-related services, and it evolved from distributed computing to the modern technology known as **cloud computing**. Cloud services include those provided by Amazon, Google, and Microsoft. Cloud computing allows users to access a wide range of services stored in the cloud or on the Internet. Cloud computing services include computer resources, data storage, apps, servers, development tools, and networking protocols.



Distributed Systems

Distributed System is a composition of multiple independent systems but all of them are depicted as a single entity to the users. The purpose of distributed systems is to share resources and also use them effectively and efficiently. Distributed systems possess characteristics such as scalability, concurrency, continuous availability, heterogeneity, and independence in failures.

But the main problem with this system was that all the systems were required to be present at the same geographical location. Thus to solve this problem, distributed computing led to three more types of computing and they were-Mainframe computing, cluster computing, and grid computing.

Mainframe Computing

Mainframes which first came into existence in 1951 are highly powerful and reliable computing machines. These are responsible for handling large data such as massive input-output operations.

Even today these are used for bulk processing tasks such as online transactions etc. These systems have almost no downtime with high fault tolerance. After distributed computing, these increased the processing capabilities of the system. But these were very expensive. To reduce this cost, cluster computing came as an alternative to mainframe technology.

Cluster Computing

In 1980s, cluster computing came as an alternative to mainframe computing. Each machine in the cluster was connected to each other by a network with high bandwidth. These were way cheaper than those mainframe systems. These were equally capable of high computations. Also, new nodes could easily be added to the cluster if it was required.

Thus, the problem of the cost was solved to some extent but the problem related to geographical restrictions still pertained. To solve this, the concept of grid computing was introduced.

Grid Computing

In 1990s, the concept of grid computing was introduced. It means that different systems were placed at entirely different geographical locations and these all were connected via the internet. These systems belonged to different organizations and thus the grid consisted of heterogeneous nodes. Although it solved some problems but new problems emerged as the distance between the nodes increased. The main problem which was encountered was the low availability of high bandwidth connectivity and with it other network associated issues. Thus. cloud computing is often referred to as "Successor of grid computing".

Virtualization

Virtualization was introduced nearly 40 years back. It refers to the process of creating a virtual layer over the hardware which allows the user to run multiple instances simultaneously on the hardware. It is a key technology used in cloud computing. It is the base on which major cloud computing services such as Amazon EC2, VMware vCloud, etc work on. Hardware virtualization is still one of the most common types of virtualization.

Web 2.0

Web 2.0 is the interface through which the cloud computing services interact with the clients. It is because of Web 2.0 that we have interactive and dynamic web pages. It also increases flexibility among web pages. Popular examples of web 2.0 include Google Maps, Facebook, Twitter, etc. Needless to say, social media is possible because of this technology only. It gained major popularity in 2004.

Service Orientation

A service orientation acts as a reference model for cloud computing. It supports low-cost, flexible, and evolvable applications. Two important concepts were introduced in this computing model. These were Quality of Service (QoS) which also includes the SLA (Service Level Agreement) and Software as a Service (SaaS).

Utility Computing

Utility Computing is a computing model that defines service provisioning techniques for services such as compute services along with other major services such as storage, infrastructure, etc which are provisioned on a pay-per-use basis.

Cloud Computing

Cloud Computing means storing and accessing the data and programs on remote servers that are hosted on the internet instead of the computer's hard drive or local server. Cloud computing is also referred to as Internet-based computing, it is a technology where the resource is provided as a service through the Internet to the user. The data that is stored can be files, images, documents, or any other storable document.

Platform as a Service | PaaS

Platform as a Service (PaaS) provides a runtime environment. It allows programmers to easily create, test, run, and deploy web applications. You can purchase these applications from a cloud service provider on a pay-as-per-use basis and access them using an Internet connection.

In PaaS, back-end scalability is managed by the cloud service provider, so end-users do not need to worry about managing the infrastructure.

PaaS includes infrastructure (servers, storage, and networking) and platform (middleware, development tools, database management systems, business intelligence, and more) to support the web application life cycle.

Examples: Google App Engine, Force.com, Joyent, Azure.

Advantages of PaaS

There are the following advantages of PaaS -

1) Simplified Development

PaaS allows developers to focus on development and innovation without worrying about infrastructure management.

2) Lower risk

No need for up-front investment in hardware and software. Developers only need a PC and an internet connection to start building applications.

3) Prebuilt business functionality

Some PaaS vendors also provide already defined business functionality so that users can avoid building everything from very scratch and hence can directly start the projects only.

4) Instant community

PaaS vendors frequently provide online communities where the developer can get ideas, share experiences, and seek advice from others.

5) Scalability

Applications deployed can scale from one to thousands of users without any changes to the applications.

Disadvantages of PaaS Loud Computing Layer

1) Vendor lock-in

One has to write the applications according to the platform provided by the PaaS vendor, so the migration of an application to another PaaS vendor would be a problem.

2) Data Privacy

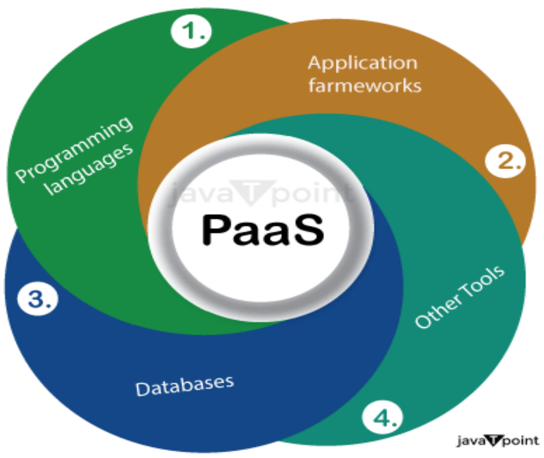
Corporate data, whether it can be critical or not, will be private, so if it is not located within the walls of the company, there can be a risk in terms of privacy of data.

3) Integration with the rest of the systems applications

It may happen that some applications are local, and some are in the cloud. So there will be chances of increased complexity when we want to use data in the cloud with the local data.

4) Limited Customization and Control: The degree of customization and control over the underlying infrastructure is constrained by PaaS platforms' frequent provision of pre-configured services and their relative rigidity.

Some of the Services Provided by PaaS are:



- **Programming Languages:** A variety of programming languages are supported by PaaS providers, allowing developers to choose their favorite language to create apps. Languages including Java, Python, Ruby,.NET, PHP, and Node.js are frequently supported.

- **Application Frameworks:** Pre-configured application frameworks are offered by PaaS platforms, which streamline the development process. These frameworks include features like libraries, APIs, and tools for quick development, laying the groundwork for creating scalable and reliable applications. Popular application frameworks include Laravel, Django, Ruby on Rails, and Spring Framework.

- **Databases:** Managed database services are provided by PaaS providers, making it simple for developers to store and retrieve data. These services support relational databases (like MySQL, PostgreSQL, and Microsoft SQL Server) and NoSQL databases (like MongoDB, Cassandra, and Redis). For its database services, PaaS platforms often offer automated backups, scalability, and monitoring tools.

- **Additional Tools and Services:** PaaS providers provide a range of extra tools and services to aid in the lifecycle of application development and deployment. These may consist of the following:
  Development Tools: to speed up the development process, these include integrated development environments (IDEs), version control systems, build and deployment tools, and debugging tools.
  Collaboration and Communication: PaaS platforms frequently come with capabilities for team collaboration, including chat services, shared repositories, and project management software.
  Analytics and Monitoring: PaaS providers may give tools for tracking application performance, examining user behavior data, and producing insights to improve application behavior and address problems.

  Security and Identity Management: PaaS systems come with built-in security features like access control, encryption, and mechanisms for authentication and authorization to protect the privacy of applications and data.

  Scalability and load balancing: PaaS services frequently offer automatic scaling capabilities that let applications allocate more resources as needed to manage a spike in traffic or demand. To improve performance and availability, load balancing features divide incoming requests among various instances of the application.

Different Computing Paradigms

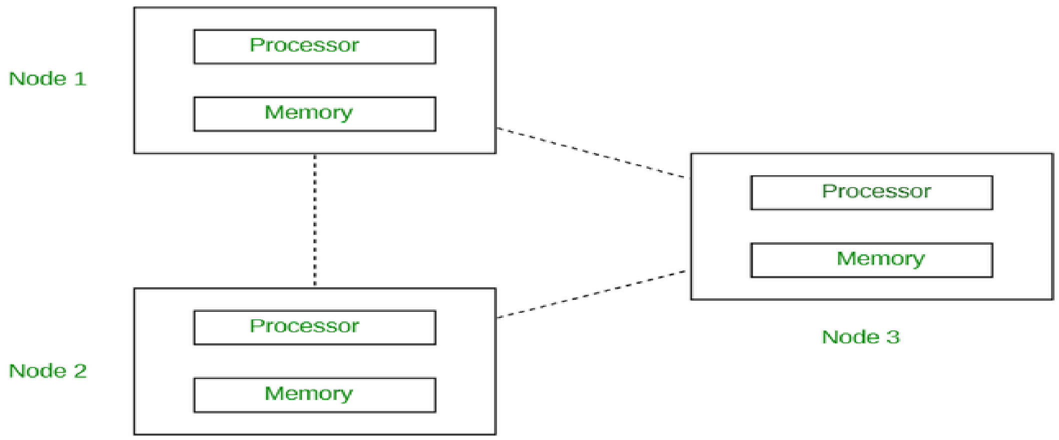
Over the years different computing paradigms have been developed and used. In fact different computing paradigms have existed before the cloud computing paradigm. Let us take a look at all the computing paradigms below.

1.Distributed Computing :

Distributed computing is defined as a type of computing where multiple computer systems work on a single problem. Here all the computer systems are linked together and the problem is divided into sub-problems where each part is solved by different computer systems.

- **The goal of distributed computing is to increase the performance and efficiency of the system and ensure fault tolerance.**

In the below diagram, each processor has its own local memory and all the processors communicate with each other over a network.



Advantages:

- **Scalability:** Can handle very large problems by distributing the workload across many machines.
- **Fault tolerance:** If one machine fails, others can continue operating.
- **Resource sharing:** Allows sharing of resources like data, software, and hardware.

Disadvantages:

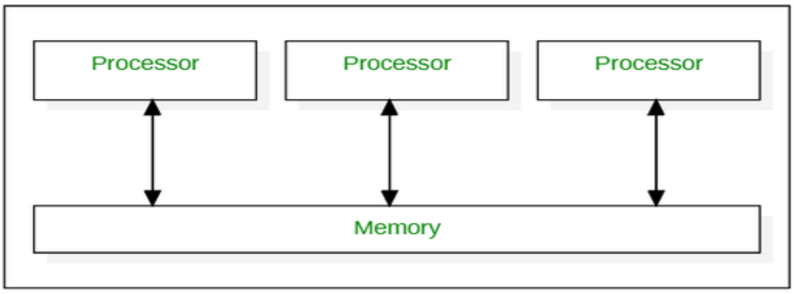
- **Complexity:** Programming is more complex due to network communication and potential latency.
- **Security:** Distributed systems are more vulnerable to security breaches.
- **Consistency:** Maintaining data consistency across multiple machines can be challenging.

2. Parallel Computing :

Parallel computing is defined as a type of computing where multiple computer systems are used simultaneously. Here a problem is broken into sub-problems and then further broken down into instructions. These instructions from each sub-problem are executed concurrently on different processors.

Here in the below diagram you can see how the **parallel computing system consists of multiple processors that communicate with each other and perform multiple tasks over a shared memory simultaneously.**

- **The goal of parallel computing is to save time and provide concurrency.**



Advantages:

- **Speedup:** Solves problems faster by dividing them into subproblems and processing them concurrently.
- **Handles larger problems:** Can tackle problems too complex for a single processor.
- **Increased throughput:** Processes more tasks in a given time.

Disadvantages:

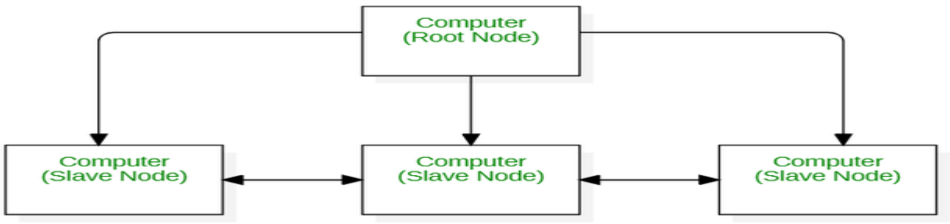
- **Complexity:** Programming can be more challenging, requiring careful management of shared resources and communication.
- **Overhead:** Communication and synchronization between processors can introduce overhead, reducing overall efficiency.
- **Scalability limitations:** Shared memory systems have limited scalability due to memory contention.

3. Cluster Computing :

A cluster is a group of independent computers that work together to perform the tasks given.
Cluster computing is defined as a type of computing that consists of two or more independent computers, referred to as nodes, that work together to execute tasks as a single machine.

- **The goal of cluster computing is to increase the performance, scalability and simplicity of the system.**

As you can see in the below diagram, all the nodes, (irrespective of whether they are a parent node or child node), act as a single entity to perform the tasks.



Advantages:

- **Cost-effective:** Uses commodity hardware, making it more affordable than specialized supercomputers.
- **Scalability:** Can be scaled by adding more nodes to the cluster.
- **High availability:** If one node fails, others can take over its tasks.

Disadvantages:

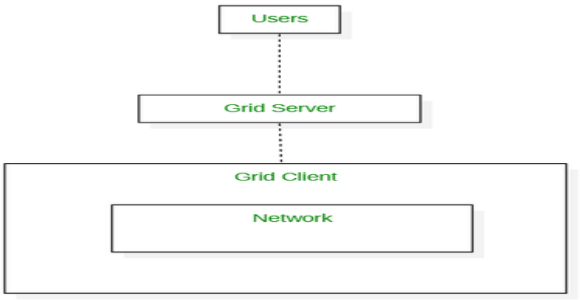
- **Management complexity:** Managing a cluster of machines can be complex.
- **Network dependence:** Performance relies heavily on the network connection between nodes.
- **Software requirements:** Requires specialized software for cluster management and job scheduling.

4. Grid Computing :

Grid computing is defined as a type of computing where it is constitutes a network of computers that work together to perform tasks that may be difficult for a single machine to handle. All the computers on that network work under the same umbrella and are termed as a virtual super computer.

The tasks they work on is of either high computing power and consist of large data sets. All communication between the computer systems in grid computing is done on the "data grid".

- **The goal of grid computing is to solve more high computational problems in less time and improve productivity.**



Advantages:

- Resource aggregation: Combines resources from geographically dispersed organizations.
- Large-scale problem solving: Enables solving very large and complex problems that require massive computing power.
- Cost-effectiveness: Leverages existing resources, reducing the need for new investments.

Disadvantages:

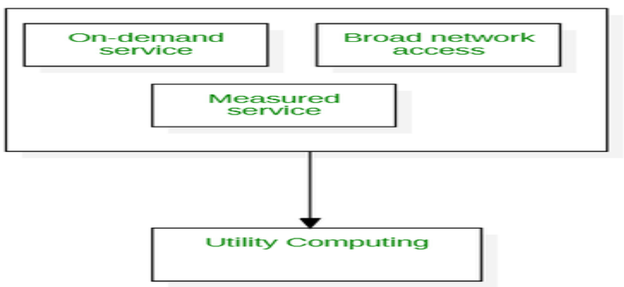
- Heterogeneity: Managing diverse hardware and software can be challenging.
- Security: Security concerns due to sharing resources across different organizations.
- Coordination: Requires complex coordination mechanisms to manage resources and jobs.

5. Utility Computing :

Utility computing is defined as the type of computing where the service provider provides the needed resources and services to the customer and charges them depending on the usage of these resources as per requirement and demand, but not of a fixed rate.

Utility computing involves the renting of resources such as hardware, software, etc. depending on the demand and the requirement.

- **The goal of utility computing is to increase the usage of resources and be more cost-efficient.**



- **Advantages:**

- Cost-effectiveness: Pay-as-you-go model, reducing upfront costs.
- Scalability: Resources can be scaled up or down as needed.
- Flexibility: Access to a wide range of computing resources.

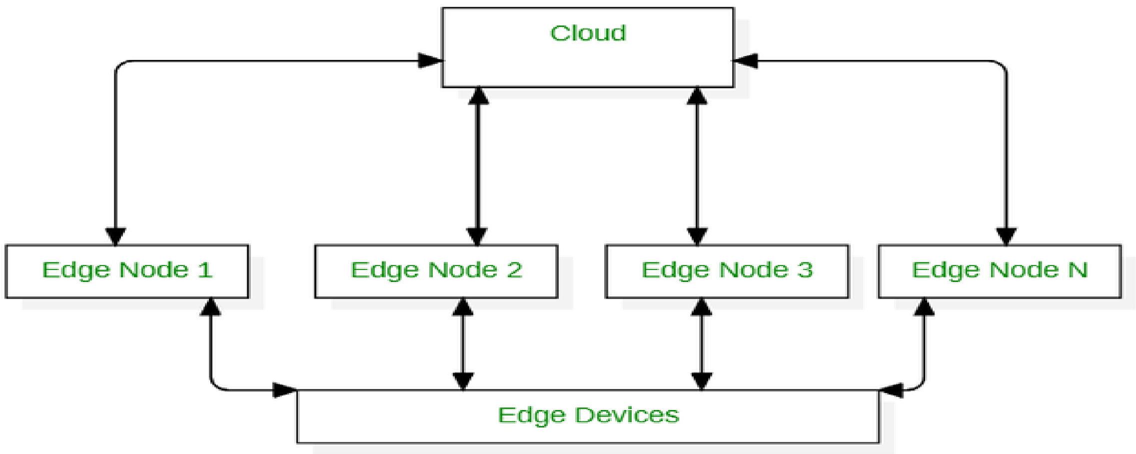
Disadvantages:

- Vendor lock-in: Dependence on a specific utility provider.
- Security and privacy: Concerns about data security and privacy in a shared environment.
- Dependence on network: Requires a reliable network connection to access resources.

6. Edge Computing :

Edge computing is defined as the type of computing that is focused on decreasing the long distance communication between the client and the server. This is done by running fewer processes in the cloud and moving these processes onto a user's computer, IoT device or edge device/server.

- **The goal of edge computing is to bring computation to the network's edge which in turn builds less gap and results in better and closer interaction.**

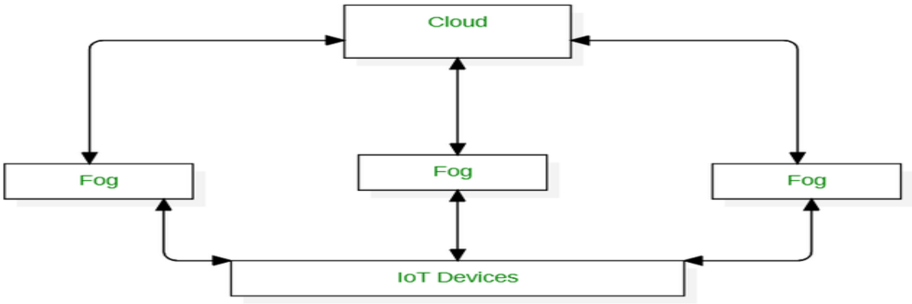


7. Fog Computing :

Fog computing is defined as the type of computing that acts a computational structure between the cloud and the data producing devices. It is also called as "fogging".

This structure enables users to allocate resources, data, applications in locations at a closer range within each other.

- **The goal of fog computing is to improve the overall network efficiency and performance**.



Advantages:

- Reduced latency: Processes data closer to the source, reducing latency.
- Bandwidth efficiency: Reduces the amount of data that needs to be transmitted to the cloud.
- Improved real-time processing: Enables faster processing of time-sensitive data.

Disadvantages:

- Management complexity: Managing a distributed fog infrastructure can be complex.
- Security: Security concerns due to the distributed nature of fog nodes.
- Resource constraints: Fog nodes may have limited resources compared to cloud data centers.

8. Cloud Computing :

Cloud computing is defined as the type of computing where it is the delivery of on-demand computing services over the internet on a pay-as-you-go basis. It is widely distributed, network-based and used for storage.